

# ZHE XUE

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## EDUCATION

Southern University of Science and Technology, China

Aug. 2023 – Jun. 2027

*B.Eng. candidate in Computer Science and Technology (Turing Class)*

- **GPA:** 3.92/4.00; **Rank:** 4/157
- **Peer Tutor, Linear Algebra** (Spring & Fall 2024): delivered exam review sessions for fellow students.

## AWARDS

**Gold Medal**, 2024 ICPC Asia Chengdu Regional Contest

Oct. 2024

**Gold Medal**, 2024 ICPC Asia Hong Kong Regional Contest

Dec. 2024

**Silver Medal**, 2024 ICPC Asia East Continent Final Contest

Dec. 2024

## RESEARCH EXPERIENCE

### Hybrid Sampling for Reliable On-Policy Distillation

May 2026 – Jul. 2026

- Contributed to the design and refinement of a likelihood-ratio-guided hybrid sampling method that selectively replaces student-proposed tokens with teacher samples before prefix deviations degrade subsequent supervision.
- Evaluated the intervention–quality trade-off, showing that 1.9–5.5% teacher-token intervention substantially improved rollout quality while preserving predominantly student-generated trajectories.
- Built a hybrid rollout pipeline on DAPO-Math-17K by adapting vLLM’s speculative-decoding engine; end-to-end training incurred 8.6% runtime overhead over OPD under a matched training-data budget.
- Distilled Qwen3-4B into Qwen3-1.7B, improving macro Avg@8 by 3.25 pp and macro Pass@8 by 4.94 pp over OPD on AIME 2024/2025 and AMC 2023.

### Uncertainty-Aware Inference for LLM Reasoning

Sep. 2025 – Present

- Characterized temperature-dependent reasoning across four Qwen3 models and five MATH difficulty levels, tracking answer accuracy, ReasonEval validity/redundancy, and Math-Shepherd verification scores.
- Observed sharp degradation beyond model- and difficulty-dependent temperature thresholds; harder problems degraded at lower temperatures.
- Benchmarked majority voting and confidence-weighted aggregation under temperature sweeps, finding that multi-sample gains peaked before high-temperature degradation.
- Investigating model-internal signals of uncertainty to guide adaptive inference for LLM reasoning.

## PROJECTS

### SysY Compiler<sup>①</sup>

Jul. 2025

- Built a compiler written in Rust for a C-like language, lowering ASTs through Koopa IR to RV32IM assembly.
- Implemented scoped name resolution, short-circuit evaluation, function calls, and multidimensional array.
- Optimized IR with 7 rewrite rules and code generation with a 13-register write-back cache.
- Passed 130 functional tests in both Koopa IR and RISC-V modes, including 20 benchmark programs.

### RISC-V CPU<sup>②</sup>

Apr. 2025 – May 2025

- Designed a five-stage pipelined RV32I CPU in Verilog supporting 38 instructions.
- Handled data/control hazards with forwarding, load-use stalls, static branch prediction, and pipeline flushing.
- Implemented UART-based instruction/data loading with a 3-stage Python deployment toolchain.
- Passed 8 Vivado simulation tests and 3 on-board assembly tests covering core modules, MMIO, and AUIPC.

## SKILLS

- **Languages:** Mandarin (Native), English (TOEFL iBT: 5.0/6.0, comparable 100/120)
- **Programming:** C++, Python, Rust, Java, Haskell, CUDA
- **Tools:** PyTorch, vLLM, Docker, Git, Bash, L<sup>A</sup>T<sub>E</sub>X